

wlk_stp2_1 Soundscape

Recording Information

Location	Near Crescent Meadow, University of Salford, UK
Coordinates	approx. 53.485974° N, -2.270827° W
Date & Time	—
Recording Equipment	Zoom F8n / Røde NT-SF1 ambisonic microphone
Recording Format	4-channel B-format (AmbiX, ACN/SN3D), 24-bit, 96 kHz

Processing & Render Details

The original 4-channel B-format ambisonic recording was decoded using a first-order UHJ stereo rendering method in REAPER 7.35. Spatial decoding used the ambix_binaural_o1 plugin with the icosahedron-3h3v preset. The raw decoded source presented an extremely low integrated loudness of -47.6 LUFS (LUFS-S -44.2; LUFS-M -42.5; True Peak -32.8 dBTP; LRA 5.0 LU), requiring a gain correction of +30.0 dB via JS: Volume Adjustment. Final limiting applied via JS: Master Limiter (ceiling -1.0 dBTP). Metered with Youlean Loudness Meter 2. Output rendered at 24-bit, 44.1 kHz, 2-channel stereo WAV (2067 kbps), 13 April 2025.

Processing Chain

DAW	REAPER 7.35
Ambisonic decoder	AmbiX VST — ambix_binaural_o1, icosahedron-3h3v preset
Gain adjustment	JS: Volume Adjustment (+30.0 dB)
Limiter	JS: Master Limiter (ceiling -1.0 dBTP)
Loudness meter	Youlean Loudness Meter 2
Render date	13 April 2025

Output Specification

Format	WAV, 24-bit PCM
Sample rate	44,100 Hz
Channels	2 (stereo)
Bit rate	2067 kbps
Duration	0:38.000

Pre-Processing Levels (raw decoded source)

Parameter	Value
Integrated Loudness (LUFS-I)	-47.6 LUFS
Short-Term Loudness (LUFS-S)	-44.2 LUFS
Momentary Loudness (LUFS-M)	-42.5 LUFS
Loudness Range (LRA)	5.0 LU
True Peak	-32.8 dBTP

Loudness Metrics

Parameter	Value
Integrated Loudness (LUFS-I)	-18.0 LUFS
Short-Term Loudness max (LUFS-S)	-14.4 LUFS
Momentary Loudness max (LUFS-M)	-12.8 LUFS
Loudness Range (LRA)	5.3 LU
True Peak	-3.8 dBTP
Clipping	0 occurrences

Loudness normalisation target: -18 LUFS ±0.5 (ITU-R BS.1770-4 / AES TD1004.1.15-10).

Rationale

Selected as part of a broader comparison strategy within the experimental dataset. Its location closely aligns with the Crescent Meadow recording, enabling perceptual comparisons across similarly situated urban greenspace environments. The large gain correction (+30.0 dB) reflects the extremely low capture level of the source material; the processing route preserved ecological realism while achieving perceptual parity with the wider stimulus set. The updated LRA of 5.3 LU confirms that useful internal dynamics were retained.

REAPER Render Analysis

REAPER Render Sun Apr 13 10:54:21 2025

WAV: 24bit PCM, 44100Hz, 2ch, 2067kbps

File	Length	Peak	Clips	max LUFS-M	max LUFS-S	LUFS-I	LRA
wlk1_stp2_0.wav	0:38.000	-3.2	0	-12.8	-14.4	-17.4	5.4

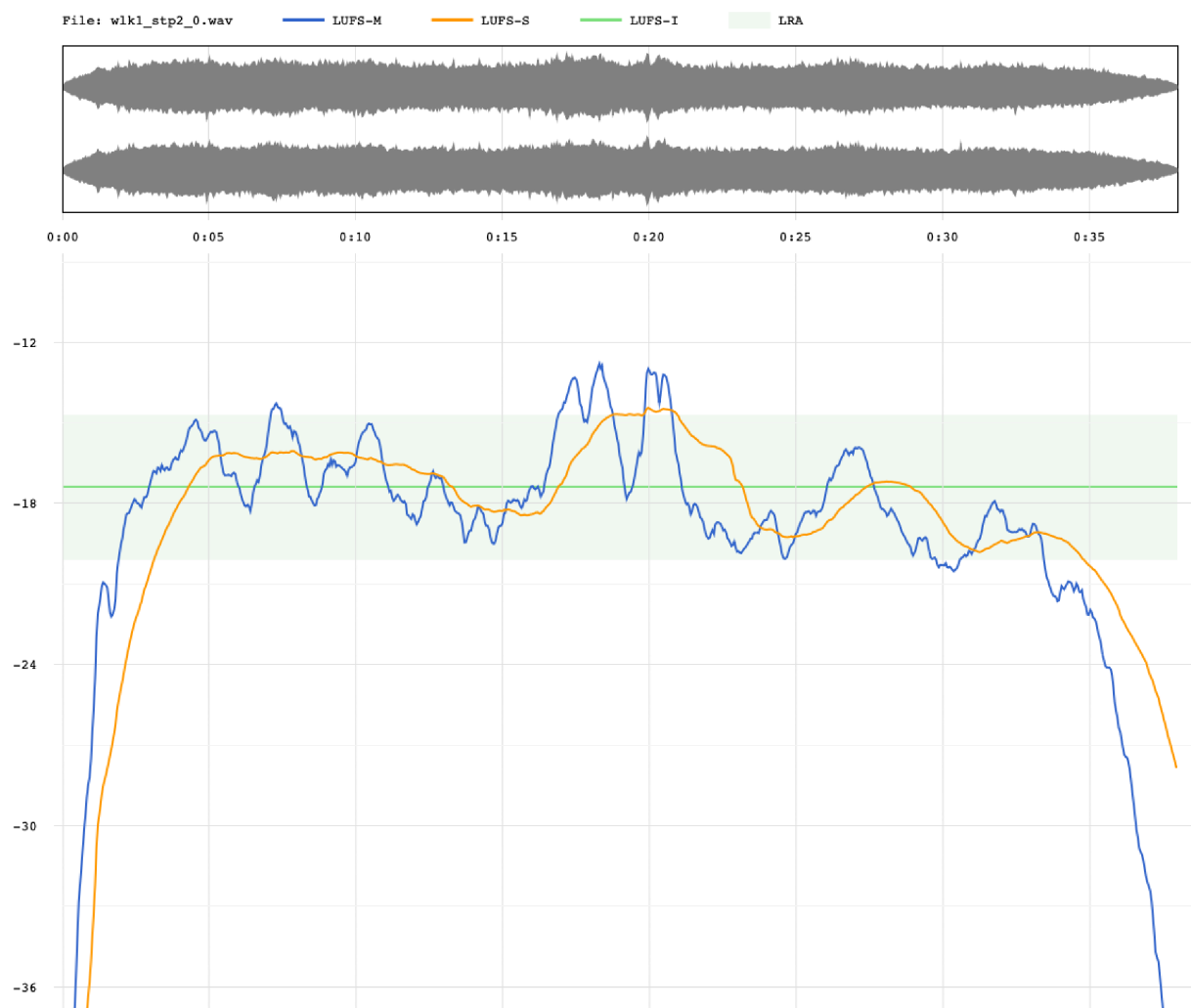


Figure: REAPER loudness render report — wlk_stp2_1 Soundscape. Waveform display (top) and integrated loudness profile (LUFS-M blue, LUFS-S orange, LUFS-I green, LRA shaded region).